CS4045D- Image Processing Course Project _Details

Topic : Geometric Augmentation

Study augmentations that change the spatial geometry of the image (position, size and orientation of objects) and measure their effect on each YOLO version's ability to localise vehicles.

Group 03: Effect of Image Scaling and Zoom Augmentation

Augmentation Factor: Scaling / Zoom

Objective: To study how training on vehicles rendered at different apparent sizes affects detection of near and distant vehicles across the three YOLO versions.

Input: Baseline dataset; chosen Low/Medium/High scale factors applied with padding/cropping to keep 640x640 output; label boxes rescaled accordingly.

Output: Scaled image sets and sample visualisations; trained models for baseline and each scale level, repeated across all 3 YOLO versions; size-stratified detection accuracy.

6-Week Plan

Week	Focus	Activities	Weekly Deliverable
Week 1	Dataset & Baseline	Prepare the assigned vehicle subset, convert annotations to YOLO format, visualise bounding boxes, create train/val/test splits, and train the baseline model with no augmentation on each of the 3 YOLO versions (YOLOv8n, YOLOv11n and YOLOv26n).	Weekly report + baseline dataset & baseline metrics for all 3 YOLO versions
Week 2	Design & Implement Assigned Augmentation	Design Low/Medium/High (or more) intensity settings for "Scaling / Zoom" and justify the choice; implement the augmentation in code; generate and visually inspect before/after sample grids for each intensity level.	Weekly report + augmentation implementation code, chosen intensity values with justification, and before/after visual samples
Week 3	Augmented Training Across 3 YOLO Versions	Train each of the 3 YOLO versions (YOLOv8n, YOLOv11n and YOLOv26n) separately on every augmented dataset version, keeping dataset split, image size, epochs, batch size and hardware identical to the corresponding baseline run.	Weekly report + trained weight files for every intensity level, across all 3 YOLO versions
Week 4	Experimental Runs & Consolidation	Consolidate results for Baseline vs each chosen intensity level, for every YOLO version; record Precision, Recall, mAP@50, mAP@50-95, training time and inference time for each run in a single comparison matrix (intensity x YOLO version).	Weekly report + consolidated experiment results matrix (intensity x YOLO version)
Week 5	Evaluation & Visualisation	Generate precision-recall curves, confusion matrix, mAP comparison charts, class-wise AP, training/validation loss curves, and sample detections (correct / false positive / false negative) for each YOLO version; analyse why the augmentation helped or hurt performance, and whether the effect differs by YOLO version.	Weekly report + evaluation visuals and written analysis across all 3 YOLO versions

Week	Focus	Activities	Weekly Deliverable
Week 6	Final Report & Presentation	Compile the full final report and prepare a short presentation summarising methodology, results and conclusions, including which YOLO version benefited most from the augmentation.	Final report + presentation slides